WORKSHEET 5.10

# Optimisation

Unit 5: Analysing Functions

## Key ideas

- Name the quantity to be made largest or smallest, and write it as a function of **one** variable, using the constraint to eliminate the other.
- State the domain: the values of the variable that make sense.
- Find the critical points, then **justify**: candidates test on a closed interval, or a sign argument over the whole domain.
- Answer the question that was asked: a dimension, a value, or a point.

## Worked examples

### Example 1 · A product

Two positive numbers have sum 20. Find the largest possible product.

**Solution.** $P(x) = x(20 - x)$, $0 < x < 20$. $P'(x) = 20 - 2x = 0$ at $x = 10$, and $P'' = -2 < 0$.

**The largest product is 100.**

### Example 2 · An open box

Squares of side $x$ cm are cut from the corners of a 12 cm by 12 cm sheet, and the sides are folded up. Find the largest possible volume.

**Solution.** $V(x) = x(12 - 2x)^2$, $0 < x < 6$. $V'(x) = (12 - 2x)(12 - 6x) = 0$ at $x = 2$ in the domain; $V \to 0$ at both ends.

$V(2) = 128$ **cm$^3$.**

### Example 3 · A nearest point

Find the points on $y = x^2$ closest to $(0, 3)$.

**Solution.** Minimise $D = x^2 + (x^2 - 3)^2$. $D' = 2x + 4x\,(x^2 - 3) = 2x\,(2x^2 - 5)$. $D$ is smaller at $x = \pm\sqrt{\frac{5}{2}}$ than at $x = 0$.

$\left(\pm\sqrt{\frac{5}{2}}, \frac{5}{2}\right)$.



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## Practice

### CORE

1. A rectangle has perimeter 40 cm. Find the largest possible area.

2. Find the smallest value of $x + \dfrac{25}{x}$ for $x > 0$.

3. Find the largest value of $8x - x^2$.

4. Why is finding where the derivative is zero not, by itself, a complete solution?

### EXAM LEVEL

5. A farmer has 80 m of fencing for three sides of a rectangular field; the fourth side is a river. Find the largest possible area.

6. A closed cylindrical can must hold $54\pi$ cm$^3$. Find the radius and height that use the least material.



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7. A rectangle has its base on the $x$-axis and its upper corners on $y = 9 - x^2$. Find the largest possible area.

8. The profit from selling $x$ items is $P(x) = -2x^2 + 120x - 400$ dollars. Find the largest possible profit.

**Question 9 · Free-response practice**

An open-topped box with a square base of side $x$ metres must have volume 108 $\text{m}^3$.

(a) Show that the surface area of the box is $S(x) = x^2 + \dfrac{432}{x}$.

(b) Find the value of $x$ that makes the surface area as small as possible.

(c) Justify that this value gives the absolute minimum, and state the dimensions and surface area of the box.



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## Worked solutions

1. $A = x(20 - x)$ is greatest at $x = 10$: 100 cm$^2$.
2. $1 - \dfrac{25}{x^2} = 0$ at $x = 5$, where the value is 10.
3. $8 - 2x = 0$ at $x = 4$: the largest value is 16.
4. A zero derivative gives a candidate only. It must be shown to be a maximum or minimum, and an absolute one.
5. $A = x(80 - 2x)$, $A' = 80 - 4x = 0$ at $x = 20$: $20 \times 40 = 800$ m$^2$.
6. $h = \dfrac{54}{r^2}$, so $S = 2\pi r^2 + \dfrac{108\pi}{r}$. $S' = 4\pi r - \dfrac{108\pi}{r^2} = 0$ at $r = 3$; then $h = 6$.
7. $A = 2x\left(9 - x^2\right)$, $A' = 18 - 6x^2 = 0$ at $x = \sqrt{3}$: $A = 2\sqrt{3}(6) = 12\sqrt{3}$.
8. $P'(x) = -4x + 120 = 0$ at $x = 30$: $P(30) = 1400$ dollars.
9. (a) With height $h$, $x^2h = 108$ so $h = \dfrac{108}{x^2}$. $S = x^2 + 4xh = x^2 + \dfrac{432}{x}$.

   (b) $S'(x) = 2x - \dfrac{432}{x^2} = 0$ gives $x^3 = 216$, so $x = 6$.

   (c) $S''(x) = 2 + \dfrac{864}{x^3} > 0$ for all $x > 0$, and $x = 6$ is the only critical point, so it is the absolute minimum. The box is $6 \times 6 \times 3$ m with surface area 108 m$^2$.



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